

Bank Guarantee Lifecycle Monitoring

Internal Financial Controls & Automated Escalation System

[Google Sheets register](#) · [Python monitoring engine](#) · [Desktop application](#) · [Management dashboard](#)

3

escalation tiers, one-directional

6

validation gates before despatch

134(5)(e)

Companies Act IFC anchor

Document control

Control attribute	Specification
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Technical architecture	Google Sheets operational register, Python monitoring engine, Windows desktop application and HTML dashboard
Authorised despatch gateway	bg-monitor@yourcompany.example (Gmail SMTP over SSL, application password)
Physical vault custodian	Manager (Finance)

Approval ledger

Role	Name & designation	Mandate	Status
Prepared by	CA Dharmendra Kumar, Senior Partner	IFC and statutory audit design	Completed
Reviewed by	Head of Internal Audit	Risk appraisal and control assurance	Reviewed
Approved by	Chief Financial Officer / Board of Directors	Executive authorisation	Approved

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Version 3.0 — what changed

This revision supersedes V2.4. The engine has been rebuilt in Python and now runs from a company workstation rather than on Google's servers; a desktop application and a management dashboard have been added; the escalation rule has been corrected so that a softer notice can no longer follow a stronger one; and the trigger has moved from hourly to a single scheduled morning run.

Sections 1, 2, 7 and 17 of V2.4 carry forward substantially unchanged.

1. Statutory mandate under Section 134(5)(e), Companies Act 2013

Section 134(5)(e) of the Companies Act, 2013 requires that the Directors' Responsibility Statement of a listed company state that the directors have laid down internal financial controls to be followed by the company, and that such internal financial controls are adequate and were operating effectively during the financial year. The statutory explanation to that sub-section defines internal financial controls as the policies and procedures adopted by the company for the orderly and efficient conduct of its business, including adherence to company policies, safeguarding of assets, prevention and detection of frauds and errors, accuracy and completeness of accounting records, and timely preparation of reliable financial information.

A bank guarantee held by the company is an asset in the sense contemplated by that provision. It is a contingent claim which is enforceable only within a defined window, and which extinguishes automatically upon expiry. A guarantee allowed to lapse without extension or invocation therefore represents an unsecured exposure that was, until the moment of lapse, fully secured. The control failure is silent: no entry is passed, no reconciliation breaks, and the loss surfaces only when recovery is attempted and found to be impossible.

Why this system exists

The monitoring described in this manual is the operating control through which the company discharges the safeguarding-of-assets limb of Section 134(5)(e) in respect of bank guarantees. It converts a custodial obligation that would otherwise depend on individual memory into an automated, evidenced and testable process.

1.1 Assertions supported

IFC assertion	How this system supports it
Safeguarding of assets	Every live guarantee is tracked to expiry and escalated in three stages, so the claim period cannot close unnoticed.
Prevention and detection of error	Six validation gates reject chronologically impossible entries at the point of capture and flag unreadable dates for correction.
Accuracy and completeness of records	The register is the single source of truth; the dashboard reports data quality exceptions such as blank expiry dates and missing vendor addresses.
Adherence to company policy	Notice wording, escalation intervals and the invocation clause are system-generated and cannot be varied by an individual operator.
Timely reliable information	The dashboard reports value at risk by month, by bank and by vendor, on demand and without manual compilation.

1.2 Evidence available to the auditor

The BG_Email_Log tab constitutes the audit trail. Each entry records a system-generated log identifier, the compound record key, the guarantee number, the expiry date, the timestamp of processing in Indian Standard Time, the despatch mode in force, the outcome, and the full text of any error returned by the mail gateway. Entries are appended and never edited. The log may be reconciled directly against the Sent Items of the authorised despatch gateway.

2. Risk exposure and mitigation of recurring audit findings

The controls below respond to findings which recur in internal and statutory audits of guarantee registers maintained manually.

Recurring audit finding	Root cause in a manual system	Automated control
Lapsed claim period	Registers not reviewed; expiry passed without notice to the vendor or to finance.	Three-tier progressive escalation at 30, 20 and 10 days, despatched to the vendor with copies to finance.
Weekend or bank holiday default	Notice served after expiry because the due date fell on a non-working day.	Rollback algorithm steps the despatch date backward to the preceding business day, never forward.
Unsecured mobilisation advance	Advance guarantee expired before contractual recovery was complete.	The 10-day notice carries an assertive invocation clause ensuring the claim is lodged within the window.
Chronological inconsistency	BG date captured earlier than the contract date, or expiry earlier than issue.	Cell-level validation enforces $H \geq E$ and $M > H$, and the engine re-tests both at every run.
Absence of an audit trail	Reminders given by telephone or from personal mailboxes, leaving no record.	Immutable BG_Email_Log records key, timestamp, mode, outcome and error text for every record examined.
Concentration risk not visible	Exposure by issuing bank or by vendor never compiled.	Dashboard reports bank-wise and vendor-wise exposure with percentage shares.

3. System architecture at a glance

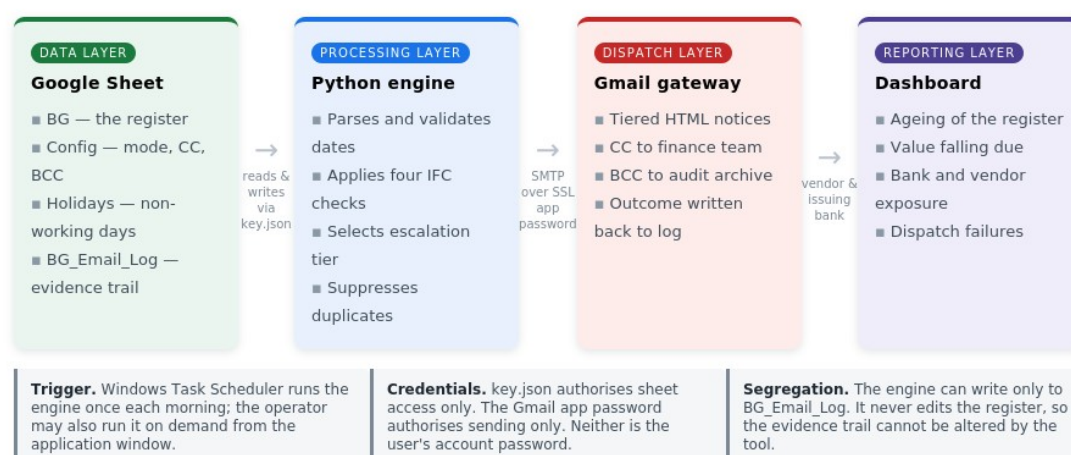


Figure 1 — Four layers: the register, the engine, the despatch gateway and the reporting view.

The register remains in Google Sheets, where the commercial and finance teams already work. The processing engine, previously written in Google Apps Script and executed on Google's servers, is now a Python program executed on a company workstation. Despatch continues through the authorised Gmail gateway. The dashboard is generated on demand as a self-contained HTML file which may be printed, emailed or attached to an audit working paper.

The one architectural trade-off

Apps Script executed on Google's infrastructure and ran whether or not any workstation was switched on. The Python engine runs on a nominated workstation, which must be powered on at the scheduled hour. The scheduled task is configured to run at the earliest opportunity after a missed start, and the dashboard reports the timestamp of the last engine run so that a prolonged outage is visible rather than silent.

This is a deliberate exchange: local execution brings the register within the company's own control environment, removes dependence on a personal Google account's trigger quota, and permits the richer reporting described in Part E.

Part B — The Register

4. Master operational register blueprint (BG sheet)

The BG tab is the sole operational repository for guarantees received. Column positions are fixed because the engine and the validation rules address them by position as well as by header text; headers must not be renamed without a corresponding change to the engine.

Col	Field header	Type	Validation and business rule
A	Sl. No.	Integer	Sequential reference index.
B	Vendor Name	Text	Registered commercial name of the contractor or supplier.
C	Recipient Email	Email	Vendor address for automated notices. If blank, notices fall back to DEFAULT_TO_EMAIL and the omission is reported as a data quality exception.
D	Contract Number	Text	Purchase order, work order or contract agreement number.
E	Contract Date	Date	Base contractual anchor date.
F	Purpose of BG	Dropdown	Mobilisation Advance / Performance Security / EMD / Retention.
G	BG Number	Text	Instrument number issued by the bank. Forms part of the

Col	Field header	Type	Validation and business rule
			deduplication key.
H	BG Date	Date	Must be on or after the Contract Date in column E.
I	Issuing Bank	Derived — do not type	Populated automatically by lookup against the RBI IFSC master from the code entered in column J. Drives bank-wise exposure reporting.
J	IFS Code	Text (entered)	Eleven-character alphanumeric code taken from the face of the guarantee. This is the only one of the three bank fields keyed by the operator.
K	Branch Details	Derived — do not type	Populated automatically by lookup against the RBI IFSC master. Carries the branch name and location.
L	BG Amount (INR)	Currency	Sanctioned sum secured. Read by the dashboard for value-at-risk reporting.
M	Date of Expiry	Date	Must be strictly later than the BG Date in column H. A blank entry removes the row from monitoring altogether.
N	Status	Dropdown	Active / Closed / Released / Invoked / Discharged / Returned / Cancelled.

4.1 Automated derivation of bank and branch from the IFS code

Columns I and K are not keyed. The operator enters only the eleven-character IFS code appearing on the face of the guarantee into column J, and the issuing bank name and branch details are returned automatically by lookup formulae against two reference tabs, IFSC-1 and IFSC-2, held within the same workbook.

Those two tabs hold the National Electronic Funds Transfer participant list published by the Reserve Bank of India at <https://www.rbi.org.in/scripts/neft.aspx>. The list is split across two tabs because a single sheet cannot conveniently hold the full national file. The lookup searches both.

Column	Source	Control effect
J — IFS Code	Keyed by the operator from the instrument	Single point of entry. A transcription error here is visible at once, because columns I and K return no match rather than a plausible but wrong bank name.
I — Issuing Bank	Derived from IFSC-1 / IFSC-2	Bank names are standardised to the RBI's own nomenclature, so bank-wise exposure in the dashboard aggregates correctly instead of splitting across spelling variants.
K — Branch Details	Derived from IFSC-1 / IFSC-2	Branch identification is authenticated against the regulator's list rather than accepted on the vendor's description.

The IFSC master is reference data and must be kept current

Bank amalgamations, branch migrations and new branch codes make the participant list a moving record. An outdated master returns no match for a validly issued recent guarantee, and continues to return a merged bank's former name for older ones.

The IFSC-1 and IFSC-2 tabs are therefore refreshed from the Reserve Bank's page at least half-yearly, and immediately following any announced amalgamation affecting a bank appearing in the register. The date of the last refresh is recorded in the Config tab and tested at internal audit.

A code that returns no match displays as an error value in columns I and K. Such rows are reported in the dashboard's data quality panel and must be resolved before the guarantee reaches its thirty-day tier — an unresolved error obscures which bank the claim would be lodged against.

A blank expiry date is not a neutral entry

A row without a date in column M is passed over in silence — it generates no notice and no exception in the log. The dashboard is the compensating control: it lists such rows under data quality exceptions. Reviewing that panel is a required step in the monthly review described at section 17.

5. Preventive controls and validation gates

5.1 Cell-level validation in the spreadsheet

Data validation rules are applied to the register through the retained Apps Script function `applySheetDataValidations()`, invoked from the BG Monitoring System menu. These operate at the point of entry:

- **Column H — BG date rule.** Formula `=H2>=E2`. An entry preceding the contract date is rejected with the message **INVALID BG DATE: BG DATE CANNOT BE EARLIER THAN THE CONTRACT DATE**.
- **Column M — expiry date rule.** Formula `=M2>H2`. An entry on or before the BG date is rejected with the message **INVALID EXPIRY DATE: DATE OF EXPIRY MUST BE LATER THAN THE BG DATE**.

5.2 Detective validation in the engine

Cell validation can be bypassed by pasting values into the sheet. The engine therefore re-applies the same tests, together with four others, at every run. A row must pass all six gates before any notice is despatched.

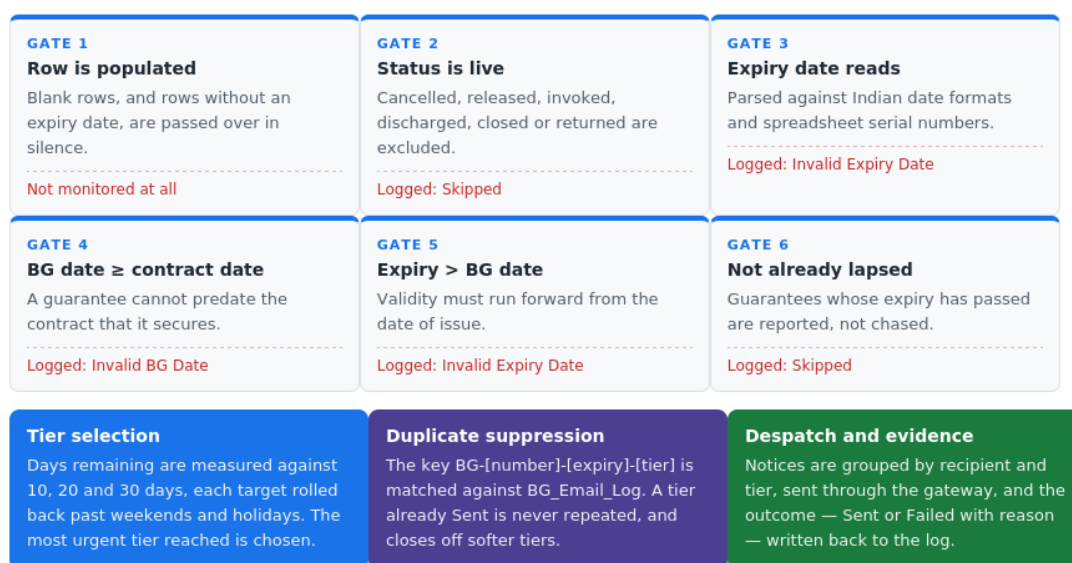


Figure 2 — The six validation gates, and what happens to a row that fails each.

Failures are not silent. Each is written to BG_Email_Log with a status describing the defect, and each appears in the dashboard's data quality panel until corrected. A defect is logged once, not at every run, so that the log remains an evidence record rather than a repetition of the same exception.

6. Three-tier escalation architecture

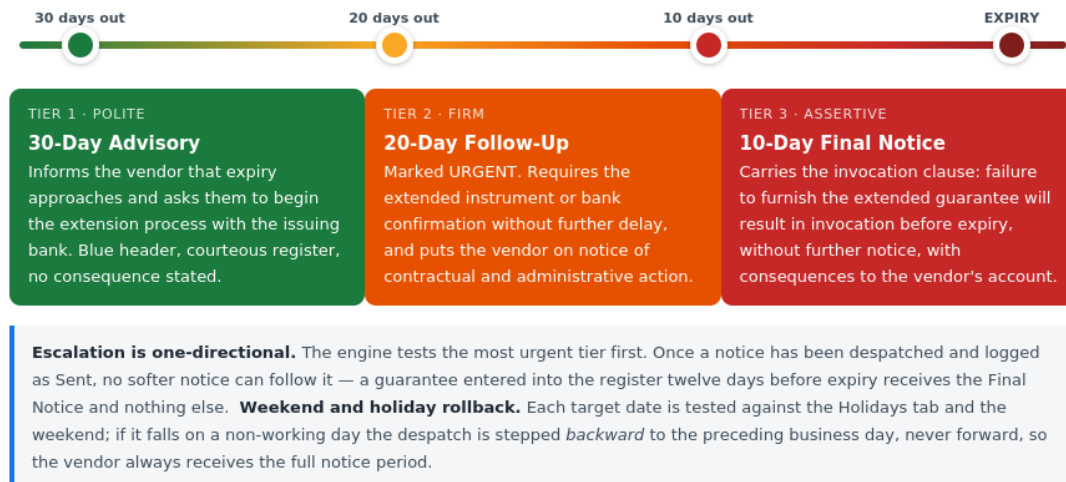


Figure 3 — Escalation intervals, tone, and the two rules governing sequence and timing.

6.1 Deduplication and sequence

For each guarantee and each tier the engine constructs a compound key in the form BG-[BG Number]-[Expiry YYYYMMDD]-[Tier]. Before despatch it tests that key against BG_Email_Log. A key already recorded as Sent is never despatched again.

The engine evaluates tiers in descending order of urgency — ten days first, then twenty, then thirty. If the most urgent tier that has fallen due has already been despatched, processing of that guarantee stops. This is a correction to the behaviour of the earlier Apps Script version, under which a guarantee entered into the register late — say twelve days before expiry — would receive the twenty-day firm notice and then, on the following day, the polite thirty-day advisory. Escalation now runs in one direction only.

6.2 Weekend and holiday rollback

The target despatch date is computed as expiry less the tier interval. That date is then tested against the Holidays tab and against Saturday and Sunday. While it falls on a non-working day it is stepped backward one day at a time until a business day is reached, so that consecutive holidays are handled correctly. Rollback is always backward, never forward, so the recipient receives no less than the stated notice period.

6.3 Grouping of notices

Records due on the same day are grouped by recipient address and by tier, so that a vendor with four guarantees reaching the thirty-day point receives one advisory listing all four rather than four separate messages. Tone is never mixed within a message: a vendor with one guarantee at ten days and another at thirty days receives two distinct notices.

Part C — Configuration & Security

7. Configuration register and access control

7.1 Config sheet

Key	Value	Operational function
SYSTEM_MODE	PRODUCTION	Routes notices to vendor addresses held in the register. Set to TEST to route every notice to the administrator's inbox with a visible test banner.
DEFAULT_TO_EMAIL	bg-monitor@yourcompany.example	Fallback recipient where column C is blank, and the sole recipient in TEST mode.
TEST_EMAIL	bg-monitor@yourcompany.example	Designated destination for system test despatches.
CC_EMAILS	[finance / commercial list]	Receives copies of all outward notices for internal visibility.
BCC_EMAILS	[audit archive inbox]	Blind copy for archiving and internal audit review.

7.2 Access control and segregation of duties

Tab or range	Protection	Authorised personnel	Control objective
Config	Protected sheet	System administrator	Prevents unauthorised change to despatch mode or recipient routing.
Holidays	Protected sheet	System administrator	Protects the holiday calendar from accidental modification.
BG_Email_Log	Protected sheet	System administrator	Preserves evidentiary entries for statutory audit verification.
IFSC-1 and IFSC-2	Protected sheets	System administrator	Protects the RBI participant list from edits that would corrupt bank and branch derivation.
BG columns I and K	Protected range	System administrator	Prevents overtyping of the lookup formulae with free text.
BG headers, row 1	Protected range	System administrator	Prevents header renaming that would break column detection.
BG data area, row 2 onward	Editable	Commercial and finance teams	Permits day-to-day entry subject to automated validation.

8. Credentials architecture and custody of secrets

Two distinct credentials are used, each with a narrow scope. Neither is the password of any individual's Google account, and neither grants access to the mailbox.

Credential	What it is	What it permits	Custody
key.json	Private key of the service account bg-bot@...iam.gserviceaccount.com, issued by Google Cloud	Read and write access to those spreadsheets which have been explicitly shared with the service account, and nothing else	Stored only in the application folder on the nominated workstation. Never emailed, never placed in a synchronised cloud folder, never transmitted in a support conversation.
Gmail app password	Sixteen-character code issued by Google under two-step verification	Sending mail through the SMTP gateway as the authorised despatch address	Held in the scheduled task's launcher file, or typed into the application window at the time of use. Revocable independently of the account password.

Compromise procedure

If key.json is exposed — by email, by upload, by loss of the workstation, or by inclusion in a screenshot or support conversation — the key must be deleted from the Keys tab of the service account in the Google Cloud console and a fresh key generated the same day. The service account itself, and every sharing permission, remains valid; only the key changes. Exposure of the app password is remedied by revoking it at myaccount.google.com and issuing another.

Neither event requires any change to the register, the engine or the schedule.

8.1 Service account authorisation chain

1. A Google Cloud project (BG MONITOR) is created under the authorised account.
2. The Google Sheets API and Google Drive API are enabled within that project.
3. A service account, bg-bot, is created and a JSON key issued and downloaded once.
4. The register is shared with the service account's address, with Editor rights, exactly as a human collaborator would be.
5. Access may be withdrawn at any time by removing that sharing permission from the sheet — no change to the workstation is required.

9. Migration from Apps Script to Python

Dimension	Google Apps Script (V2.4)	Python engine (V3.0)
Execution	Google's servers, independent of any workstation	Nominated company workstation, under Task Scheduler
Availability	Ran regardless of local hardware	Requires the workstation to be powered on at the scheduled hour
Frequency	Hourly between 08:00 and 18:00 IST	Once each working morning; on demand at any time
Escalation sequence	Softer notice could follow a stronger one	One-directional; corrected
Exception logging	Repeated at every run, inflating the log	Recorded once per defect
Reporting	None beyond the log	Management dashboard with exposure and ageing analysis
User interface	Spreadsheet menu and web app	Desktop application with preview, test, live and dashboard actions
Portability	Bound to the Google account	Standalone executable deployable to any Windows machine

9.1 What was deliberately retained in Apps Script

The function `applySheetDataValidations()` remains in the spreadsheet and continues to enforce the cell-level rules described at section 5.1, since those must operate at the point of entry inside Google Sheets. The Apps Script time-based trigger and web app deployment have been disabled; both systems write to the same log with the same key format, so an accidental overlap would be suppressed, but running both remains prohibited.

10. One-time installation and authorisation

Step	Action	Verification
1	Install Python 3.9 or later on the nominated workstation, selecting Add python.exe to PATH during setup.	python --version returns a version number at the command prompt.
2	Install the supporting libraries: pip install gspread google-auth tzdata	The installation ends with a Successfully installed message.
3	Create the Google Cloud project and enable the Sheets and Drive APIs.	Both APIs show status Enabled in the API library.
4	Create the bg-bot service account and download a JSON key; save it into the application folder as key.json.	The folder shows key.json with file extensions displayed.
5	Share the register with the service account address, with Editor rights.	The service account appears in the sheet's Share dialogue.
6	Record the spreadsheet identifier in the engine (the portion of the sheet's address between /d/ and /edit).	A preview run reads the register without a permission error.
7	Enable two-step verification on the despatch account and generate a sixteen-character app password.	A test despatch reaches the administrator's inbox.
8	Disable the Apps Script time-based trigger and archive any web app deployment.	The Triggers page in the Apps Script editor is empty.

Display file extensions before you begin

Windows hides file extensions by default, which causes a downloaded key to be renamed key.json.json without any visible sign. In File Explorer, open View, then Show, and tick File name extensions before handling either credential file.

11. Folder layout and file inventory

File	Purpose	Contains a secret?
bg_monitor.py	The monitoring engine — validation, tier selection, despatch and logging	No (holds the spreadsheet identifier only)
bg_dashboard.py	Builds the management dashboard as an HTML file	No
bg_gui.py	The desktop application window	No
key.json	Service account private key	Yes — restricted custody
BG_Monitor.exe	Console build of the engine, for scheduled runs	No
BG_Monitor_App.exe	Windowed build of the desktop application	No
daily_bg_run.bat	Launcher used by Task Scheduler	Yes — holds the app password
bg_run.log	Local record of each scheduled run's console output	No

Keep the application folder outside OneDrive

A folder inside OneDrive is replicated to Microsoft's cloud and to every device signed in to the same account. Both files marked above as holding a secret would travel with it. The recommended location is a local path such as C:\BG_App, excluded from synchronisation and covered by the workstation's disk encryption policy.

12. Operating the system — three modes

Mode	Command	What it does	When to use it
Preview	python bg_monitor.py --preview	Reads the register, reports which notices are due, sends nothing and writes nothing to the log.	Before every live run; whenever the register has been edited substantially.
Test despatch	python bg_monitor.py --test	Routes every notice to the administrator's inbox with a test banner, whatever the Config setting.	After any change to wording, configuration or the register's structure.
Live	python bg_monitor.py	Processes and despatches according to SYSTEM_MODE in the Config tab, and writes the outcome to the log.	The scheduled morning run, and manual runs when a guarantee is added close to expiry.

Preview is not optional before going live

The ten-day notice states that the guarantee will be invoked without further reference to the vendor. A mistyped expiry date therefore does not produce a harmless early reminder — it produces a legal warning to a counterparty whose guarantee is in order. The preview list must be read against the physical instruments before the first live run following any bulk entry.

13. The desktop application

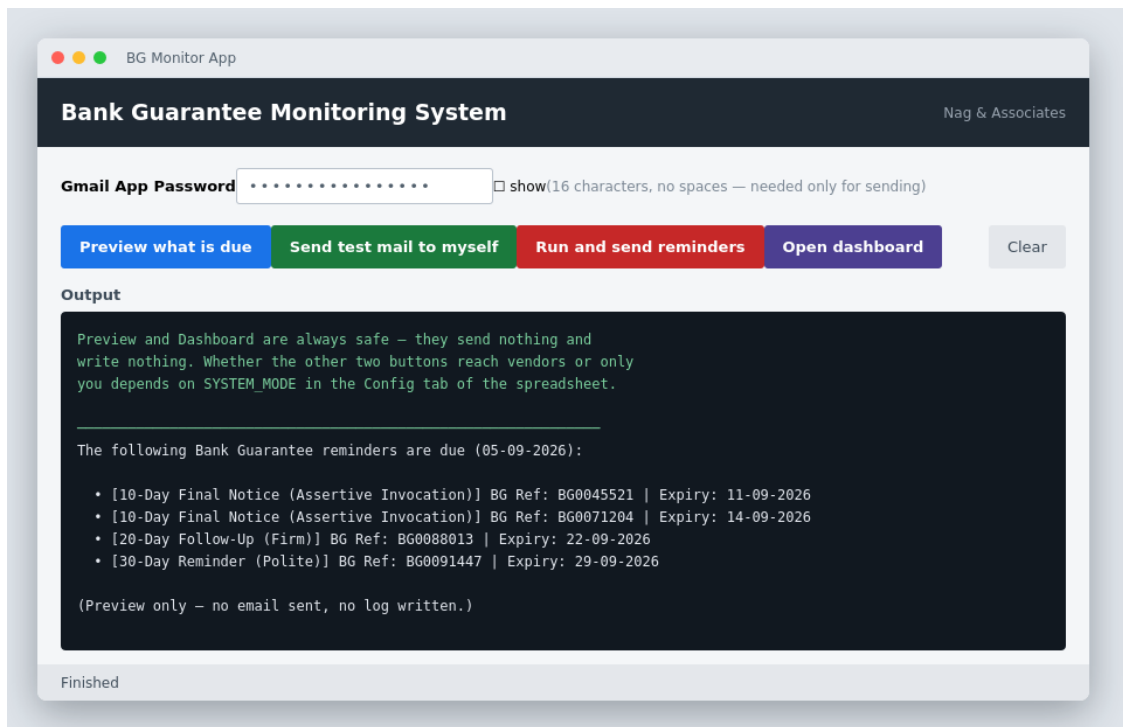


Figure 4 — The application window, showing a preview of notices due.

Control	Effect	Safe?
Preview what is due	Lists the notices that would be despatched today. No mail, no log entry.	Always safe
Send test mail to myself	Despatches to the administrator only, with the test banner, regardless of Config.	Safe
Run and send reminders	The live run. Confirmation is requested before anything is despatched.	Reaches vendors
Open dashboard	Builds the dashboard and opens it in the browser. Read-only.	Always safe
Gmail App Password field	Accepts the sixteen-character code. Required only for the two despatch actions.	Not stored to disk

The password field is empty at every launch by design, so that the credential is never written to the workstation's disk by the application. The output panel reproduces exactly what the engine reports, including the reason for any failure, and may be copied into an incident record.

14. Building the standalone executable

The executable allows the tool to be deployed to a workstation on which Python is not installed. It is built from the same source and behaves identically.

1. Install the builder once: `pip install pyinstaller`
2. Build the console engine: `python -m PyInstaller --onefile --name BG_Monitor bg_monitor_exe.py`
3. Build the desktop application: `python -m PyInstaller --onefile --windowed --name BG_Monitor_App bg_gui.py`
4. Copy the two executables from the dist folder into the deployment folder, together with `key.json`.
5. Delete the build folder and the `.spec` files, which are intermediate artefacts.

Two behaviours to expect

A one-file executable unpacks itself to a temporary folder at each launch, so the first window may take ten to twenty seconds to appear. SmartScreen will warn on first execution because the file is unsigned; this is characteristic of all unsigned executables and is not evidence of a defect.

The executable embeds the spreadsheet identifier and the escalation logic. Any change to intervals, wording or the register's location requires the source to be edited and the executable rebuilt — it does not update itself.

15. Daily automation through Task Scheduler

Setting	Value
Task name	BG Monitor — daily run
Trigger	Daily, 09:30 hours
Action	Start a program: C:\BG_App\daily_bg_run.bat
Settings	Run task as soon as possible after a scheduled start is missed
Output	Appended to <code>bg_run.log</code> in the application folder
Review	The dashboard reports the timestamp of the last engine run; a stale timestamp indicates a missed schedule.

A single daily run is sufficient. Escalation intervals are measured in whole days, so a guarantee entering a tier at any hour is captured at the following morning's run, and the duplicate suppression rule means that additional runs during the day would find nothing to do.

Part E — Management Dashboard

16. Reading the dashboard

The dashboard is generated on demand from the live register. It writes a self-contained HTML file which opens in the default browser, changes nothing in the spreadsheet, and may be printed to PDF for circulation or for attachment to an audit working paper.

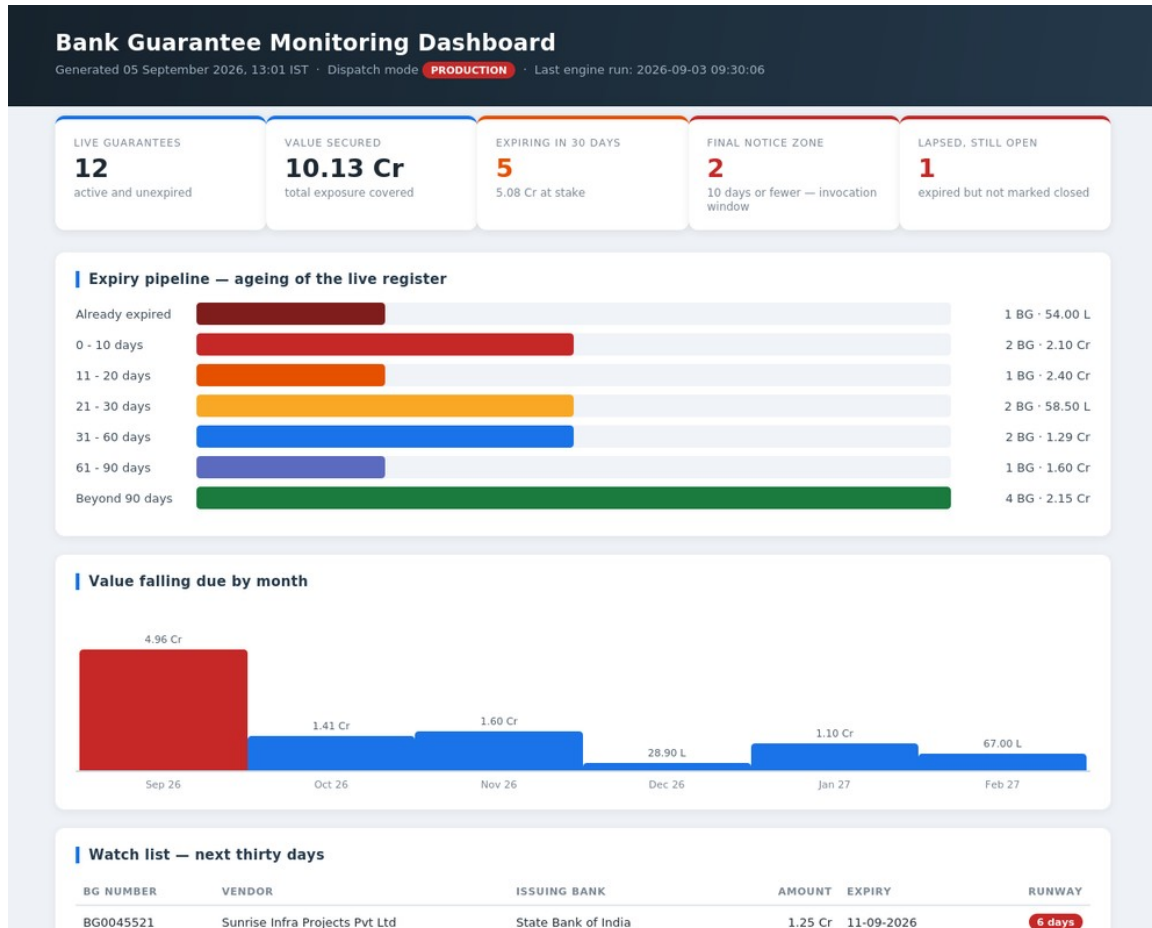


Figure 5 — Headline position, ageing of the register, and value falling due by month.

Panel	What it answers	Control use
Headline tiles	How many guarantees are live, what value they secure, how many are within thirty and within ten days, and how many have lapsed while still marked open.	The lapsed-but-open figure should be nil. Any figure above nil is a register hygiene failure requiring same-day correction.
Expiry pipeline	The ageing of the live register across seven buckets from lapsed to beyond ninety days.	Establishes whether renewal effort is concentrated in the near term or spread evenly.
Value falling due	Rupee value reaching expiry in each of the next six months.	Supports treasury planning and identifies months of concentrated renewal load.

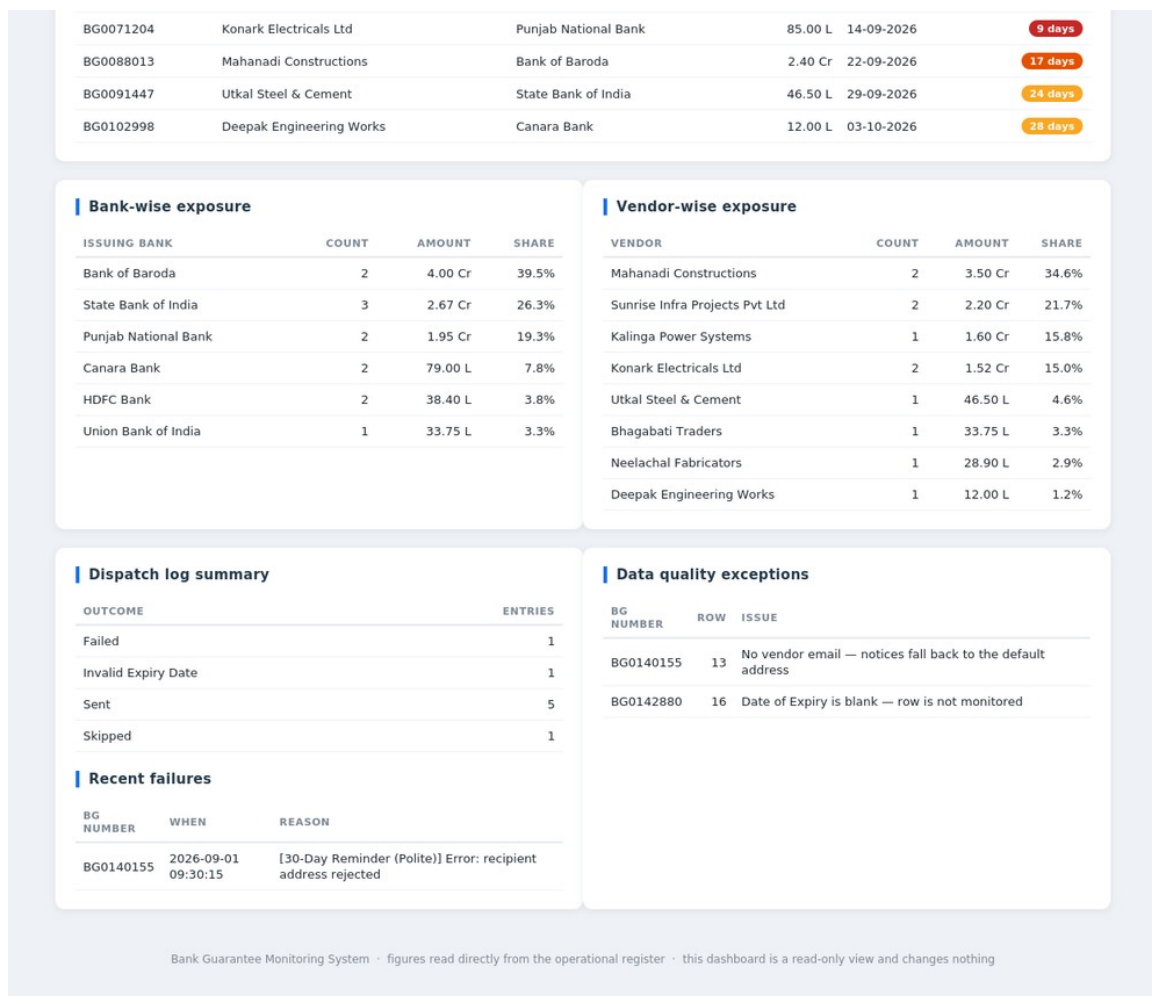


Figure 6 — Watch list, concentration analysis, dispatch outcomes and data quality exceptions.

Panel	What it answers	Control use
Watch list	Every guarantee expiring within thirty days, ordered by runway, with the vendor, bank and amount.	The working agenda for the commercial team. Cross-check against physical instruments held in the vault.
Bank-wise exposure	Value secured through each issuing bank, with percentage share.	Identifies concentration with a single issuer, relevant to counterparty risk assessment.
Vendor-wise exposure	Value held against each contractor, with percentage share.	Supports contract review and identifies vendors whose failure would leave the largest gap.
Dispatch log summary	Counts of Sent, Failed, Skipped and Invalid entries in the log.	A rising Failed count indicates a gateway or address problem requiring investigation.
Recent failures	The last eight despatch failures with reason.	Each failure represents a notice the vendor did not receive and must be followed up manually.
Data quality exceptions	Rows with blank or unreadable expiry dates, missing vendor addresses, or impossible date sequences.	Should be cleared to nil at each monthly review. Rows listed here are not being monitored.

16.1 The dashboard as a single page

Reproduced below in full, as the reviewer sees it on opening. Printed to PDF at A4 it occupies one page, which makes it convenient as a standing agenda item for the monthly review at section 17.2 and as an attachment to the audit working paper for the guarantee cycle.

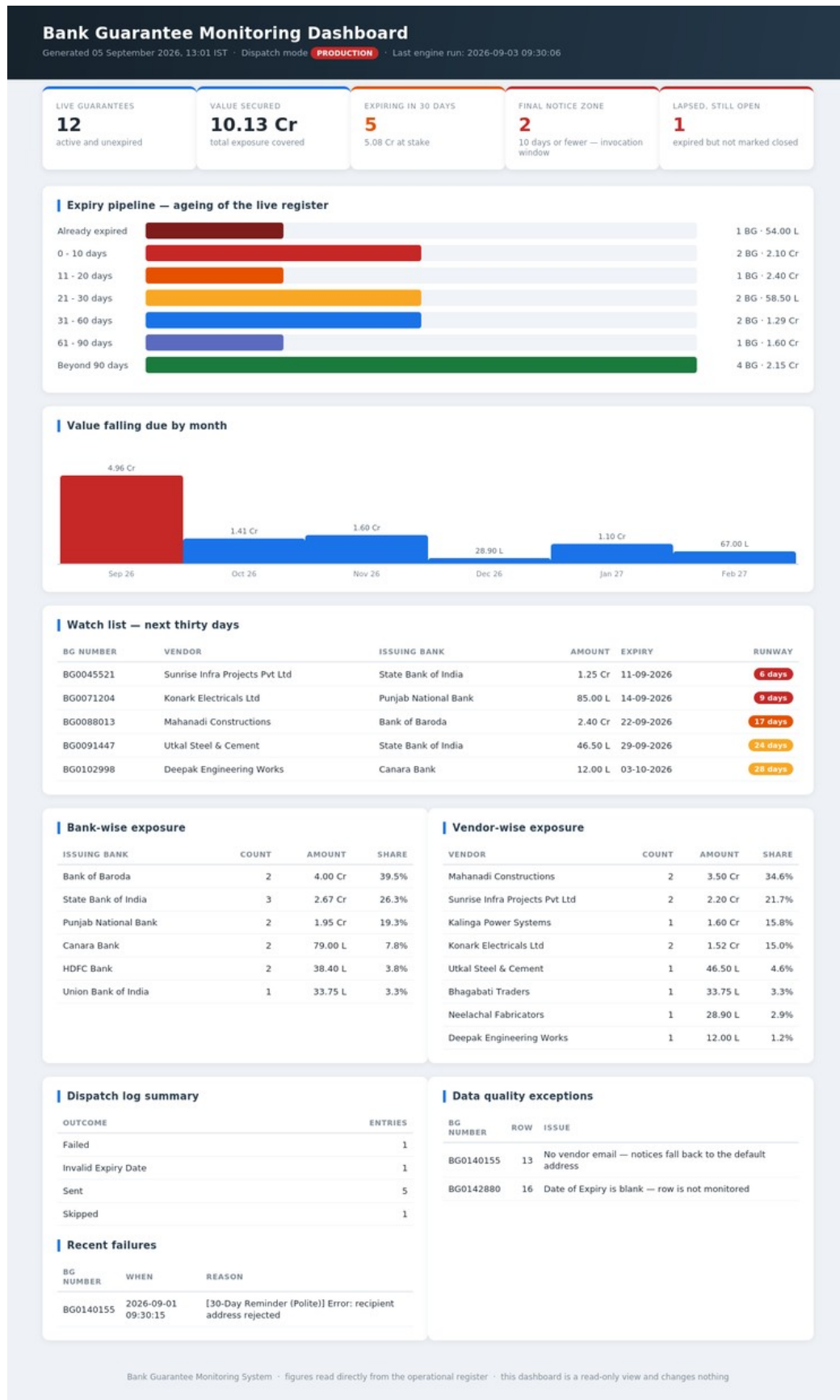


Figure 7 — The complete dashboard as generated, showing all eight panels in one view.

17. Day-to-day workflow and vault custody

1. Receive the original instrument. The commercial team receives the stamped physical guarantee from the contractor together with the bank's advice letter.
2. Verify through SFMS. Confirm that the issuing bank has transmitted the Structured Financial Messaging System message (760COV / 767COV) and obtain confirmation from the company's operating bank.
3. Record in the register. Enter the contract details, contract date, BG date and expiry date. Validation rejects chronologically impossible entries at the point of entry.
4. Hand over for custody. The original is handed to the Manager (Finance) as designated vault custodian and logged into the fireproof safe register with its packet reference.
5. Automated tracking follows. The scheduled morning run despatches the thirty, twenty and ten-day notices as expiry approaches, and records each despatch in the log.
6. Act on the final notice. On receipt of a ten-day alert the Manager (Finance) pulls the physical packet and prepares the invocation claim letter, so that lodgement is possible before the claim period closes.
7. Close the record. On extension, the register is updated with the revised expiry date, which generates a fresh escalation cycle. On return or discharge, the status is changed so that the guarantee leaves the monitored population.

17.1 Physical custody standards

- All original guarantees, extension letters and SFMS confirmations are stored in numbered fireproof treasury lockers under the custody of the Manager (Finance).
- No physical instrument may be released, discharged or returned to a contractor without joint written sign-off from the Project Head and the Head of Finance.
- The register's live population is reconciled to the contents of the safe half-yearly, and the reconciliation is retained as audit evidence.

17.2 Monthly review

At the beginning of each month the system administrator opens the dashboard and confirms: that the last engine run is recent; that no guarantee appears as lapsed while still open; that the data quality panel is empty; and that no despatch failures remain unaddressed. The review takes a few minutes and is the principal compensating control over the automation.

18. Internal audit compliance checklist (IFC testing)

Audit area	Verification procedure	Frequency	Pass criterion
Chronological validation	Attempt to enter a BG date earlier than the contract date, and an expiry on or before the BG date.	Quarterly	Both entries are rejected with the prescribed message.
Engine-side validation	Paste an invalid date pair directly into the register, bypassing cell validation, and run a preview.	Quarterly	The row is excluded and an exception is written to the log.
Evidentiary despatch trail	Select a sample of Sent entries from BG_Email_Log and trace to the Sent Items of the despatch gateway.	Quarterly	Complete agreement between log entries and gateway despatches.
Escalation sequence	Enter a test guarantee expiring in twelve days and run a preview on two consecutive days.	Half-yearly	The final notice tier is selected; no softer notice follows on the second day.
IFSC master currency	Inspect the refresh date recorded in Config and compare a sample of codes in column J against the Reserve Bank's current published list.	Half-yearly	Master refreshed within the last six months; no unresolved error values in columns I or K.
Holiday rollback	Identify guarantees whose computed target date falls on a Sunday or a listed holiday.	Half-yearly	Despatch is shifted backward to the preceding business day.
Invocation clause integrity	Review outward ten-day notices in the gateway's sent records.	Half-yearly	The mandatory invocation clause appears in full, without omission or alteration.
Schedule operation	Inspect the last engine run timestamp on the dashboard and the contents of bg_run.log.	Monthly	Runs are present for every working day, or exceptions are explained.
Credential custody	Inspect the application folder location and confirm exclusion from cloud synchronisation.	Half-yearly	key.json and the launcher reside on local storage only; key age is within policy.
Physical custody	Reconcile the live population in the register against the contents of the safe.	Half-yearly	Complete agreement; every live guarantee is physically present.
Segregation of duties	Attempt to edit the Config and BG_Email_Log tabs using an operator account.	Half-yearly	Both are protected against the operator.

19. Troubleshooting

Symptom	Likely cause	Resolution
Robot key file not found	key.json absent, or named key.json.json because extensions were hidden.	Display file extensions and rename to key.json in the application folder.
Permission denied when reading the sheet	The register has not been shared with the service account.	Share the sheet with the bg-bot address, granting Editor rights.
Username and password not accepted	App password mistyped, or generated before two-step verification was enabled.	Generate a fresh app password and retype it without spaces.
No pending reminders when notices are expected	The tier was already despatched and logged as Sent, or the expiry column	Inspect BG_Email_Log for the guarantee; confirm the column M header text is

Symptom	Likely cause	Resolution
	was not detected.	unchanged.
Notices sent twice	The Apps Script trigger is still enabled alongside the scheduled task.	Disable the trigger in the Apps Script editor and confirm the Triggers page is empty.
Scheduled task did not run	Workstation powered off at the scheduled hour.	Confirm the missed-start setting; run manually from the application window the same day.
Bank or branch shows an error value in column I or K	The IFS code is mistyped, or the branch is not present in the IFSC master held in the workbook.	Re-check the code against the instrument; if correct, refresh IFSC-1 and IFSC-2 from the Reserve Bank's NEFT page.
Bank-wise exposure omits a guarantee	The derived bank name is an error value, so the dashboard groups it under not stated.	Resolve the IFSC lookup; the figure corrects itself at the next dashboard build.
Dashboard shows a stale last-run timestamp	The scheduled task has been failing silently.	Inspect bg_run.log for the error text and resolve before the next escalation date.

20. Change control

Any change to escalation intervals, notice wording, the invocation clause, or the identity of the despatch gateway requires the approval of the Chief Financial Officer, recorded in the approval ledger of this manual, before the source is amended and the executables rebuilt. Following any such change the system is exercised in test mode before live despatch resumes, and the version number of this document is incremented.

End of document · SOP/FIN-COMM/BG-MON/2026/V3.0 · Confidential

Annexure A — Retained Apps Script listing

The migration to Python at section 9 retained one function inside the spreadsheet. Cell-level validation must operate at the moment of entry, inside Google Sheets, and cannot be performed by an external program. The listing below is reproduced in full so that the control described at section 5.1 can be verified without opening the script editor.

It is invoked from the spreadsheet menu: BG Monitoring System, then Apply Data Validations (Cols E, H, M). It is run once at implementation and again whenever rows are added beyond the validated range.

```
/**
 * Apply data validation rules to columns H and M of the BG sheet.
 * Retained from V2.4. No other Apps Script function remains active.
 */
function applySheetDataValidations() {
  const ss = SpreadsheetApp.getActiveSpreadsheet();
  const bgSheet = ss.getSheetByName('BG');
  if (!bgSheet) {
    SpreadsheetApp.getUi().alert("Sheet 'BG' not found.");
    return;
  }

  const maxRows = Math.max(bgSheet.getMaxRows(), 1000);

  // Column H – BG Date must not precede the Contract Date in column E
  const bgDateRule = SpreadsheetApp.newDataValidation()
    .requireFormulaSatisfied('=H2>=E2')
    .setHelpText('INVALID BG DATE: BG DATE CANNOT BE EARLIER THAN THE '
      + 'CONTRACT DATE.')
    .setAllowInvalid(false)
    .build();
  bgSheet.getRange(2, 8, maxRows - 1, 1).setDataValidation(bgDateRule);

  // Column M – Date of Expiry must be strictly later than the BG Date
  const expiryDateRule = SpreadsheetApp.newDataValidation()
    .requireFormulaSatisfied('=M2>H2')
    .setHelpText('INVALID EXPIRY DATE: DATE OF EXPIRY MUST BE LATER '
      + 'THAN THE BG DATE.')
    .setAllowInvalid(false)
    .build();
  bgSheet.getRange(2, 13, maxRows - 1, 1).setDataValidation(expiryDateRule);

  SpreadsheetApp.getUi().alert('Data validations applied to Columns H '
    + 'and M across all rows.');
```

Everything else in the Apps Script project is dormant

The time-based trigger, the web application deployment and the processing functions of V2.4 have been disabled as required by section 9.1. The code may remain in the project, but the Triggers page must be empty and no deployment may be active. This is tested at internal audit under the schedule operation head of section 18.

Annexure B — Python source files

The source is attached to this document as embedded objects rather than printed. Printed listings run to some fifteen pages, cannot be executed, and are superseded the moment a line is amended; an embedded file can be extracted, inspected and run, and its version is fixed at the date of embedding recorded in Annexure C.

How the files are attached

1. Place `bg_monitor.py`, `bg_monitor_exe.py`, `bg_gui.py` and `bg_dashboard.py` into a single folder and compress it to `BG_Source_V3.zip`. Mail gateways block loose `.py` files; a compressed archive survives transmission.
2. In this document, click at the marked insertion point below.
3. On the ribbon choose Insert, then Object, then Create from File.
4. Browse to `BG_Source_V3.zip`, tick Display as icon, and click OK.
5. Record the embedding date and the file hash, if one is taken, in Annexure C.

INSERTION POINT — EMBEDDED SOURCE ARCHIVE

Insert `BG_Source_V3.zip` here as an object displayed as an icon

Two files are never attached to this document

`key.json` is the service account private key. `daily_bg_run.bat` contains the Gmail application password in plain text. Neither may be embedded in this manual, circulated by mail, or placed in any synchronised folder. Both are held only on the nominated workstation, as required by section 8.

Where the launcher must be documented, it is reproduced with the credential line shown as: set
`BG_GMAIL_APP_PASSWORD=<held separately by the system administrator>`

Annexure C — Source custody and version register

This register fixes which version of each artefact was in force when this manual was approved. It is the primary evidence tested when an auditor asks whether the program that ran on a given date is the program described in this document.

Artefact	Function	Version / date embedded	Master copy held by	Approval reference
bg_monitor.py	Monitoring engine — validation, tier selection, despatch, logging	V3.0 — _____	System administrator	_____
bg_monitor_exe.py	Engine variant built into the executables	V3.0 — _____	System administrator	_____
bg_gui.py	Desktop application window	V3.0 — _____	System administrator	_____
bg_dashboard.py	Management dashboard generator	V3.0 — _____	System administrator	_____
applySheetDataValidations()	Cell-level validation retained in the spreadsheet	V2.4 — carried forward	System administrator	Annexure A
IFSC-1 / IFSC-2	RBI NEFT participant list supporting bank derivation	Refreshed _____	System administrator	Section 4.1
BG_Monitor.exe	Console build used by the scheduled task	Built _____	System administrator	Section 14
BG_Monitor_App.exe	Windowed build used by the operator	Built _____	System administrator	Section 14

Certification

The undersigned certify that the source embedded at Annexure B is the source from which the executables in production were built, that no undocumented amendment has been made since the approval date recorded above, and that the credentials excluded under section 8 are held in the custody stated.

	System administrator	Head of Internal Audit	Chief Financial Officer
Signature			
Name			
Date			

When this register must be updated

Any rebuild of an executable, any amendment to a source file, and any refresh of the IFSC master requires a fresh entry in this register before the change is placed in production. An entry made after the event is not evidence of control; it is a reconstruction.